

Dashboard — OpenDesign token vendoring + dark mode — QA evidence

Revision: 2 **Last modified:** 2026-07-09T23:45:00Z **Scope:** dashboard/ only (React 18 + Vite 5, no Tailwind). Executes docs/research/ opendesign_vendoring_plan_20260709/VENDORING_PLAN.md §2 — vendor design-systems/helix-ota/tokens.css, repoint inline hex → semantic var(--token), add a real light/dark theme, and re-prove §11.4.170 over BOTH themes. Does NOT touch clients/ota-manager/, design-systems/, or the server. **Authority:** consumer-side QA evidence (§11.4.35). Anti-bluff §11.4 — every claim below cites a captured artifact under docs/ qa/20260709-dashboard-hostrender/ (the harness’s evidence home).

1. Files changed (dashboard scope)

File	Change
src/styles/tokens.css	NEW — verbatim copy of design-systems/helix-ota/tokens.css (184 lines, provenance header preserved; zero value edits).
src/theme.ts	NEW — applyTheme/initTheme/toggleTheme/currentTheme: writes explicit data-theme on <html> + localStorage("helix-ota-dash-theme"), seeded from prefers-color-scheme.
src/main.tsx	import "./styles/tokens.css" + import { initTheme }; initTheme() called BEFORE createRoot().render() (neutralizes UNCONFIRMED R4 auto-dark by always setting an explicit data-theme).

File	Change
src/components/ui.tsx	All inline hex in styles + badgeTone → var(--token) per the plan §2.3-step-3 mapping table (see §2).
src/components/AppShell.tsx	shell bg/fg → var(--bg)/var(--fg); header kept as fixed brand chrome, toned at border via var(--border); theme toggle button added to the header user cluster (☾ Dark / ☼ Light).
src/screens/LoginScreen.tsx	session-expired notice #854d0e → var(--warn).
src/components/ui.test.tsx	§11.4.120 reconciliation — 3 Badge value-equality tests updated from old hardcoded rgb() to the NEW var(--token)/color-mix() mechanism (jsdom supplement, NOT a substitute for the host-render pixel proof).
hostrender/ login.hostrender.spec.ts	Parametrized over ["light", "dark"]; per-theme goldens; light ↔ dark distinctness test added; golden-bad baseline-reject test reworked to be --update-snapshots-safe (see §5).
hostrender/*-snapshots/login-card-light-chromium-linux.png	Regenerated 400×242 (border #e2e5ea → var(--border)=#e2e8f0 is a real pixel delta ⇒ new golden IS the evidence, §11.4.170).
hostrender/*-snapshots/login-card-dark-chromium-linux.png	NEW dark golden 400×242.

2. Hex → token mapping (faithful to the plan §2.3 table)

ui.tsx: card bg #fff → var(--surface), card/th/input/btnSecondary border #e2e5ea/#cbd2dc → var(--border), card radius 8 → var(--radius-md), btn radius 6 → var(--radius-sm), btnPrimary #2563eb/#fff → var(--accent)/var(--accent-on), fieldLabel/th/empty #4b5563/#6b7280 → var(--

muted), input bg/fg explicit var(--surface)/var(--fg), progressTrack #eef1f5 → var(--surface-warm), progressFill #2563eb → var(--accent), errorPanel #fef2f2/#fca5a5/#991b1b → color-mix(in oklab, var(--danger), transparent 92%/55%)/var(--danger), badge tones → var(--surface-warm)/color-mix(... var(--success|warn|danger|accent) ...). AppShell.tsx: shell #f5f7fa/#111827 → var(--bg)/var(--fg). LoginScreen.tsx: #854d0e → var(--warn).

3. Build result — CLEAN

```
cd dashboard && npm run build → EXIT 0. tsc --noEmit (3 projects)
clean + vite build OK. tokens.css bundled: dist/assets/index-*.css
2.02 kB. ✓ 48 modules transformed. ✓ built in 752ms.
```

4. Unit suite — GREEN (no regression)

```
cd dashboard && npm run test:run → EXIT 0, Test Files 12 passed
(12), Tests 107 passed (107). The 3 Badge value-equality tests were
§11.4.120-reconciled to the token mechanism (fix-breaks-its-own-gate:
reconciled, not reverted, not fake-passed — the “err distinct from ok”
test still asserts a real distinction).
```

Controller coverage added (§11.4.134 review-finding closure). The independent §11.4.142 review flagged that theme.ts — the load-bearing dark-mode controller — had ZERO automated coverage (the host-render harness sets data-theme directly, bypassing it). Closed with two new files, both real assertions (not tautologies): - src/theme.test.tsx (13 jsdom tests) — applyTheme writes exact data-theme + localStorage("helix-ota-dash-theme") value + a throw-tolerant path (mocked setItem throws → still sets attribute, no throw); initTheme seed precedence (persisted > OS prefers-color-scheme > light) + corrupt-value + matchMedia-absent fallthroughs; toggleTheme flips attribute+storage both directions; currentTheme reflects the attribute. Each FAILS if the controller logic breaks. - src/components/AppShell.test.tsx (1 RTL test) — renders real AppShell, clicks the theme toggle, asserts <html data-theme> flips light → dark → light + the button’s accessible label re-renders. (@testing-library/react already a devDependency — no new dep added.) Determinism re-run of the two new files: 14 passed again.

5. §11.4.170 host-render dual-oracle — BOTH themes PASS

cd dashboard && npm run e2e:hostrender (clean verdict pass, no --update-snapshots) → **9 passed, EXIT 0**. Per theme: golden-good toHaveScreenshot, layout oracle (good-PASS + mutated-FAIL-detected), pixelmatch analyzer self-check, committed golden-baseline-rejects-mutated; plus one light ↔ dark distinctness test.

Oracle (captured artifact)	light	dark
golden-good toHaveScreenshot vs committed 400×242 baseline	PASS	PASS
layout oracle good (layout-oracle-good-<t>.json)	pass:true, 0 failures	pass:true, 0 failures
layout oracle mutated FAIL-detected (layout-oracle-bad-<t>.json)	hidden-title + overlap caught	hidden-title + overlap caught
pixelmatch self-check (image-diff-selfcheck-<t>.json) good ↔ good	ratio 0.000	ratio 0.000
pixelmatch self-check good ↔ mutated	ratio 0.130 (>0.01)	ratio 0.131 (>0.01)
committed golden rejects mutated (baseline-rejects-mutated-<t>.json)	dims 242 → 175 differ	dims 242 → 175 differ

Self-validation verdict (both): SELF-VALIDATED: analyzer passes golden-good AND flags golden-bad — the analyzers PROVABLY cannot bluff (§11.4.107(10)).

--update-snapshots clobber bug found + fixed (§11.4.102 systematic-debug)

First update run produced 400×175 goldens (wrong; the true card is 242). Root cause: under --update-snapshots the golden-bad “REJECTS a mutated render” test also called `toHaveScreenshot(login-card-<t>.png)`, which in update mode WRITES the baseline — from the MUTATED render (h2 hidden + button pulled out of flow ≈ 175px) — clobbering the good 242px golden. Fix: that golden-bad test now reads the committed baseline from disk and compares EXPLICITLY (never calls `toHaveScreenshot`), so it can never be an update target. Re-update → correct 400×242 goldens; verdict pass → all 9 green.

6. Dark mode WORKS — pixel proof (distinct dark render)

- **Distinctness** (`light-vs-dark-distinctness.json`): the SAME /login screen differs by **93,070 / 921,600 px = 10.1%** of the viewport between light and dark → DISTINCT: dark is a genuine re-themed surface (not a no-op/recolor).
- **Actual card-surface pixel** (`login-<t>-actual.png`, sampled @ (8,8)):
 - light card surface = `rgb(255,255,255) = #ffffff` (= --surface light)
 - dark card surface = `rgb(2,8,23) = #020817` (= --surface dark)
Exactly the token values — the dark surface renders genuinely dark, driven by `<html data-theme="dark">` flipping `var(--surface)` with zero per-component logic. This is the §11.4.170 device-independent host-rendered-pixel proof that dark mode is real and working.

7. Honest boundary (§11.4.6)

The host-render harness sets `data-theme` directly (the same authoritative attribute the app’s `initTheme/toggleTheme` write) — it proves the token layer + both rendered themes are correct and distinct. The controller that produces that attribute in the running app (`theme.ts`) and the AppShell toggle BUTTON wiring (`click → toggleTheme → attribute flip`) are now BOTH asserted by automated tests (`src/theme.test.tsx + src/components/AppShell.test.tsx`, §4) — the earlier

gap is closed. Two proof layers now compose: jsdom tests prove the controller/DOM/storage logic, and the §11.4.170 host-render proves the resulting pixels in both themes.

Sources verified

- Files read/edited 2026-07-09 from the working tree (paths cited inline).
- No external service docs required (pure in-repo CSS/React vendoring).